

Tribhuvan University

Faculty of Humanities and Social Sciences

Machine Learning-Based Prediction of Academic Stress Among Undergraduate Students

A Project Report

Submitted To:

Department of Computer Application

Xavier International College, Boudha, Kathmandu

In partial fulfillment of the requirements for the Bachelor in Computer Application

Submitted By:

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SUPERVISOR'S RECOMMENDATION

I hereby recommend that this project is prepared under my supervision by **Santosh Shrestha** and **Saroj Lamichhane** entitled **"Machine Learning-Based Prediction of Academic Stress Among Undergraduate Students"** in partial fulfillment of the requirements for the degree of Bachelor of Computer Application recommended for the final evaluation.

……

Mr. Sulav Shrestha

Project Supervisor

Department of Computer Application

Xavier International College



Tribhuvan University

Faculty of Humanities and Social Sciences

Xavier International College

LETTER OF APPROVAL

This is to certify that this project is prepared by **Santosh Shrestha** and **Saroj Lamichhane** entitled **"Machine Learning-Based Prediction of Academic Stress Among Undergraduate Students"** in partial fulfillment of the requirements for the degree of Bachelor of Computer Application has been evaluated. In our opinion, it is satisfactory in the scope and quality as a project for the required degree.

…… Mr. Sulav Shrestha Project Supervisor Department of Computer Application Xavier International College	…… Mr. Roshan Jha HOD Department of Computer Application Xavier International College
…… Internal Examiner	…… External Examiner

ACKNOWLEDGEMENT

We would like to express our deepest appreciation and sincere gratitude to everyone who supported us in completing the **"Machine Learning-Based Prediction of Academic Stress Among Undergraduate Students in Nepal"** project. Their guidance, encouragement, and support were invaluable throughout this journey.

First and foremost, we extend our heartfelt thanks to our project supervisor, **Mr. Sulav Shrestha**, for his continuous guidance, valuable suggestions, and academic support throughout the planning, implementation, evaluation, and documentation of this project. His expertise and direction greatly contributed to its successful completion.

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We also acknowledge the open-source communities and developers of **Python, pandas, NumPy, scikit-learn, XGBoost, SHAP, Jupyter, and Streamlit** for providing the tools and technologies used in our research prototype.

Thank you all for your valuable contributions, guidance, and support towards the successful completion of our project.

Yours Sincerely,

Saroj Lamichhane
Santosh Shrestha

ABSTRACT

Academic stress among undergraduate students may be influenced by academic demands, demographic characteristics, lifestyle patterns, and psychosocial conditions. This project develops a survey-based Machine Learning research prototype to investigate whether such information can be used to classify undergraduate students into Low, Moderate, and High academic stress categories. Data are collected through an anonymous voluntary questionnaire containing demographic, academic, lifestyle, psychosocial, and ten stress-assessment variables. Selected stress items are reverse-scored and combined into a total stress score, after which percentile-based thresholds are used to construct the three-class target.

The implemented workflow covers data understanding, preprocessing, exploratory data analysis, target construction, feature engineering, stratified train-test splitting, five-fold stratified cross-validation, model comparison, held-out testing, SHAP-based explainability, and error analysis. Seven classifiers are evaluated with a Dummy Classifier baseline. The current dataset contains 39 observations and 37 final model features. The Decision Tree records the highest cross-validation Macro F1-score (0.562 ± 0.262), while its held-out test Macro F1-score is 0.267 with 0.250 accuracy on eight test observations. A constrained Decision Tree using maximum depth 3 and minimum samples per leaf 3 improves the same split to 0.357 Macro F1 and 0.375 accuracy. Because the dataset and test set are very small, these results are preliminary. The Streamlit application is presented only as a research demonstration and is not intended for diagnosis or high-stakes decisions.

Keywords: Academic Stress, Machine Learning, Classification, Undergraduate Students, Survey Data, SHAP, Explainable AI, Streamlit.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
BCA	Bachelor of Computer Application
CGPA	Cumulative Grade Point Average
CV	Cross-Validation
DT	Decision Tree
EDA	Exploratory Data Analysis
F1	F1-Score
HOD	Head of Department
KNN	K-Nearest Neighbors
ML	Machine Learning
SHAP	SHapley Additive exPlanations
SVM	Support Vector Machine
UML	Unified Modeling Language
XGBoost	Extreme Gradient Boosting

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CHAPTER 1
INTRODUCTION

1.1 Introduction

Academic stress is a common challenge experienced by undergraduate students as they manage academic requirements, examinations, assignments, attendance, future career expectations, financial concerns, and personal responsibilities. The level of stress experienced by students may be influenced by a combination of academic, demographic, lifestyle, and psychosocial factors.

With the development of data-driven techniques, Machine Learning (ML) can be used to investigate relationships and patterns within collected student data. Classification algorithms can be applied to categorize observations into different stress-level groups based on selected characteristics.

This project, titled “Machine Learning-Based Prediction of Academic Stress Among Undergraduate Students in Nepal,” focuses on developing a survey-based Machine Learning research prototype. The project investigates whether demographic, academic, lifestyle, and psychosocial characteristics can be used to classify undergraduate students into three academic stress categories: Low, Moderate, and High.

1.2 Problem Statement

Undergraduate students experience academic pressure from multiple sources, including study requirements, examinations, assignments, attendance, career concerns, financial conditions, lifestyle patterns, and social factors. Academic stress is therefore not represented by a single factor and may involve interactions among several characteristics.

The central problem addressed by the project is: How can survey-based demographic, academic, lifestyle, and psychosocial information of undergraduate students be processed. and analyzed using Machine Learning techniques to investigate patterns associated with Low, Moderate, and High levels of academic stress?

The current dataset contains only 39 observations and 37 final model features, while the held-out test set contains eight observations. Therefore, the implementation is not intended to provide a generalizable or clinically reliable prediction of academic stress.

Instead, it demonstrates a complete Machine Learning workflow and identifies limitations requiring future data collection and validation.

1.3 Objectives

1.3.1 General Objective

The general objective of this project is to develop a Machine Learning-based research prototype for investigating and classifying academic stress levels among undergraduate students in Nepal using survey-based demographic, academic, lifestyle, and psychosocial information.

1.3.2 Specific Objectives

1. To collect and preprocess anonymous survey data and prepare the dataset for analysis.
2. To construct a three-level academic stress target and perform exploratory analysis of relevant demographic, academic, lifestyle, and psychosocial attributes.
3. To train and compare multiple Machine Learning classification algorithms using stratified cross-validation and held-out testing.
4. To interpret model behavior using SHAP and error analysis and demonstrate the resulting workflow through a Streamlit research prototype.

1.4 Scope and Limitations

1.4.1 Scope

Table 1.1 Project Scope

Scope Area	**Included Work**
Data Collection	Anonymous voluntary Google Form-based academic stress survey.
Data Preprocessing	Data understanding, cleaning, transformation, encoding, and preparation.
Stress Classification	Stress score from ten questions and three categories: Low, Moderate, High.
EDA	Descriptive and graphical analysis of collected variables and stress scores.

Scope Area	Included Work
Machine Learning	Seven classifiers plus a Dummy Classifier baseline.
Model Evaluation	Stratified train-test split, five-fold cross-validation, accuracy, Macro F1, and held-out testing.
Explainability	SHAP-based exploratory feature contribution analysis.
Prototype Application	Streamlit research interface demonstrating the classification workflow.

1.4.2 Limitations

- The current dataset contains only 39 observations, which is insufficient for strong population-level conclusions.
- The final feature matrix contains 37 features for 39 observations, giving a feature-to-sample ratio of approximately 0.95.
- Only eight observations are in the held-out test set; one prediction changes accuracy by 12.5 percentage points.
- The sample is not sufficiently large or diverse to establish representation of undergraduate students across Nepal.
- The data are self-reported and may contain response and measurement biases.
- No independent external dataset is used for validation.
- SHAP results are exploratory and may be unstable because of the small dataset and weak held-out generalization.
- The Streamlit application is a research prototype and should not be used as a diagnostic or high-stakes decision tool.

1.5 Development Methodology

The project follows an Agile and iterative development approach in which planning, data collection, preprocessing, exploratory analysis, model development, evaluation, prototype development, and documentation are performed in repeated cycles. The methodology supports progressive refinement because findings from validation and error analysis can be used to improve later stages. The approach keeps data processing, model training, evaluation, explainability, and application development as separate but connected activities.

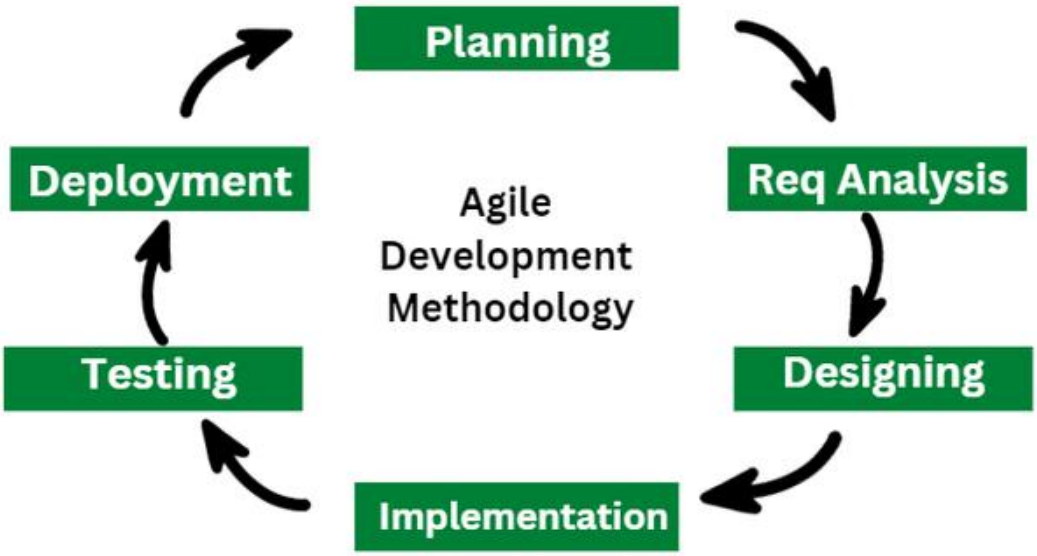


Figure 1.1 Agile / Iterative Development Methodology of Academic Stress Prediction

1.6 Report Organization

Chapter 1: Introduction

It introduces the project, problem, objectives, scope, and limitations.

Chapter 2: Background Study and Literature Review

It presents the theoretical background and literature related to academic stress, Machine Learning, preprocessing, model evaluation, explainability, and related studies.

Chapter 3: System Analysis and Design

It presents system analysis and design, including requirements, feasibility, architecture, data flow, algorithms, modules, and privacy design.

Chapter 4: Implementation and Testing

It describes implementation and testing, including the development environment, preprocessing, feature engineering, model training, results, explainability, error analysis, and prototype application.

Chapter 5: Conclusion and Future Recommendation

It summarizes the work, discusses the conclusions supported by the current evidence, and presents future enhancements.

CHAPTER 2
BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

This chapter presents the fundamental concepts and existing research related to academic stress among undergraduate students and the application of Machine Learning techniques for stress detection and classification. The purpose is to establish the theoretical foundation for the methodology adopted in this project.

2.1.1 Academic Stress

Academic stress refers to stress experienced in relation to academic requirements and educational activities. University students may encounter pressure from workload, examinations, performance expectations, financial conditions, social relationships, lifestyle patterns, and uncertainty about future careers. Recent reviews indicate that student stress is multidimensional and may involve personal, psychological, social, environmental, and coping-related factors [1], [2].

2.1.2 Factors Associated with Student Stress

Research has examined sleep, financial difficulties, social support, academic demands, personal expectations, coping mechanisms, and environmental conditions as factors associated with perceived stress. A 2025 systematic review and meta-analysis of 26 studies involving 7,828 university students in Africa reported significant associations between perceived stress and sleep problems and financial difficulties [3].

The present project therefore includes variables such as sleep hours, study hours, attendance, assignment stress, examination count, social media usage, physical activity, screen time, financial stress, social support, career stress, academic life satisfaction, academic performance, thought break, and demographic characteristics. These variables are treated as candidate predictive features rather than assumed causal factors.

2.1.3 Machine Learning

Machine Learning is a computational approach in which algorithms learn patterns from data and use those patterns to make predictions or classifications. In supervised learning, input variables are associated with known target values during training. Classification is a supervised learning task in which the target consists of discrete classes.

The present project formulates academic stress prediction as a multiclass classification problem with Low, Moderate, and High target classes.

2.1.4 Classification Algorithms

Table 2.1 Classification Algorithms Used

Algorithm	**Description**
Decision Tree	Recursively divides observations using feature-based decision rules. It is interpretable but can overfit when unconstrained.
Logistic Regression	Linear/probabilistic classification method used as a baseline-style approach.
Random Forest	Ensemble of decision trees designed to represent nonlinear relationships and reduce dependence on a single tree.
KNN	Classifies an observation using the classes of nearby observations; the project uses k=5.
XGBoost	Gradient-boosting framework based on sequentially constructed decision trees.
SVM	Finds decision boundaries between classes; the project uses an RBF kernel.
Naive Bayes	Probabilistic classifier based on Bayes' theorem and a conditional-independence assumption.

2.2 Literature Review

A literature review is a summary of previous research on a topic. The studies related to student stress detection and Machine Learning that were considered for this project are discussed below.

Schaab et al. (2024) conducted a systematic review examining Machine Learning models for detecting depression, anxiety, and stress among undergraduate students. The review analyzed multiple studies and emphasized that reported performance must be interpreted in light of methodological quality and validation practices. Most models used survey-based data, and sample size significantly affected model reliability. The review recommended larger datasets and external validation for future research. [1]

Abd-alrazaq et al. (2024) performed a systematic review and meta-analysis examining wearable AI for stress detection among students. The study included 19 studies and found that performance varied with factors such as dataset size, number of stress classes, and ground-truth method. The authors recommended using wearable AI alongside other assessments until stronger evidence is available. [2]

A 2025 systematic review of 26 studies involving 7,828 university students in Africa reported significant associations between perceived stress and sleep problems and financial difficulties. The study found that stress is multidimensional and involves personal, psychological, social, and environmental factors. [3]

A literature review on academic stress detection in classroom settings found that psychological self-questionnaires were common, followed by biological signal measurements and pattern-recognition methods. The review reported that questionnaire-based approaches are widely used because they capture subjective experiences. [4]

Lundberg and Lee (2017) introduced SHAP as a unified framework for interpreting model predictions. SHAP provides feature-attribution values that help explain why a model made a particular prediction. This study provides the theoretical foundation for explainability analysis in the present project. [5]

Pedregosa et al. (2011) developed scikit-learn, a Machine Learning library in Python that provides tools for data analysis and modeling. The library includes classification, regression, and clustering algorithms. The present project uses scikit-learn for implementing classification algorithms. [6]

Breiman (2001) introduced Random Forests, an ensemble learning method that constructs multiple decision trees and combines their predictions. The method represents nonlinear relationships and reduces dependence on a single tree. This algorithm is included in the present project. [7]

Cortes and Vapnik (1995) introduced Support Vector Networks, a classification method that finds decision boundaries between classes. The method uses kernel functions to handle nonlinear relationships. The present project uses SVM with an RBF kernel. [8]

Chen and Guestrin (2016) developed XGBoost, a scalable tree boosting system. The method uses gradient boosting with sequentially constructed decision trees. XGBoost is known for its efficiency and performance in structured data classification. The present project includes XGBoost as one of the classification algorithms. [10]

Cover and Hart (1967) introduced the Nearest Neighbor pattern classification method, which classifies observations based on the classes of nearby observations. K-Nearest Neighbors has been widely used as a baseline method. The present project uses KNN with k=5 for stress classification. [9]

CHAPTER 3
SYSTEM ANALYSIS AND DESIGN

3.1 System Analysis

This chapter describes the analysis and design of the Machine Learning-based academic stress prediction research prototype. It presents system requirements, feasibility analysis, system architecture, data flow, target construction, feature engineering, model design, algorithms, modules, security and privacy, and reproducibility.

The proposed system is a research-oriented Machine Learning workflow. It accepts student survey responses, prepares the dataset, constructs an academic stress target, trains and compares classification models, evaluates generalization, and provides explainability and error analysis.

3.1.1 Requirement Analysis

Requirement analysis identifies what the proposed system must do and the qualities it must have. The following requirements are considered.

i. Functional Requirement

1. Accept survey data collected through the academic stress questionnaire.
2. Inspect dataset dimensions, variable types, missing values, and data characteristics.
3. Clean and validate survey responses.
4. Calculate the stress score from ten stress-assessment questions.
5. Construct Low, Moderate, and High stress categories.
6. Perform exploratory data analysis.
7. Encode ordinal and categorical variables.
8. Prevent target leakage by removing target-derived variables.
9. Train multiple classification models.
10. Perform cross-validation and held-out testing.
11. Compare model performance with a Dummy Classifier baseline.
12. Generate SHAP-based feature contribution analysis.

13. Perform error and prediction-confidence analysis.
14. Provide a Streamlit research prototype.
15. Store relevant research outputs such as metrics, figures, and model artifacts.

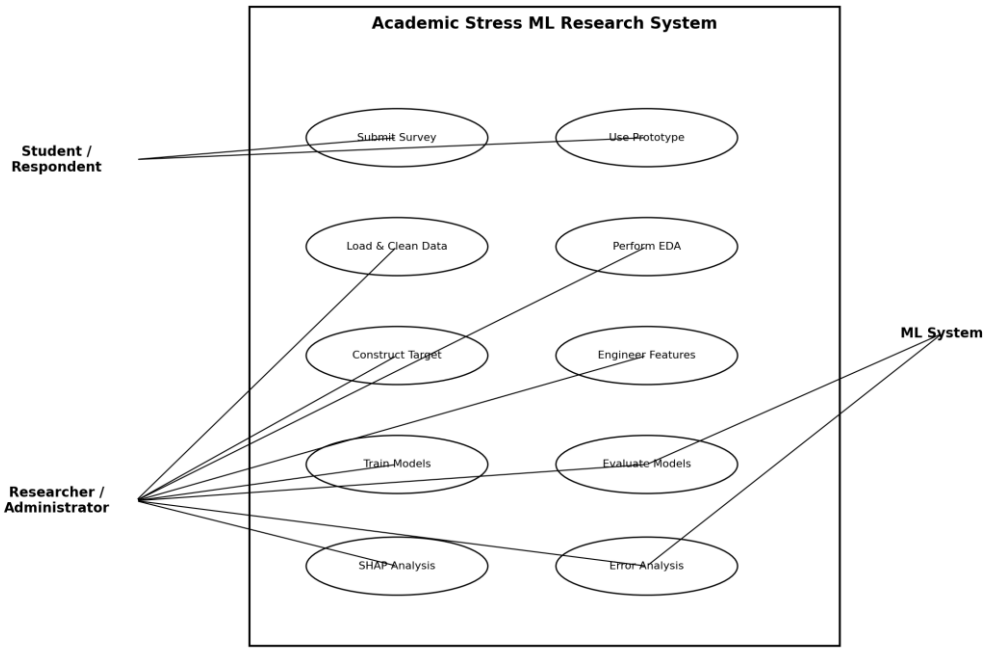


Figure 3.1 Use Case Diagram of Academic Stress Prediction

The use case diagram above shows the interaction of the system with its external actors. The student submits the anonymous survey and may use the Streamlit prototype, while the researcher performs the data and modelling tasks: loading and cleaning the dataset, performing exploratory data analysis, constructing the target, engineering features, training and evaluating the models, and performing SHAP and error analysis.

ii. Non-Functional Requirement

Table 3.1 Non-Functional Requirements

Requirement	Description
Usability	The prototype should present an understandable workflow and prediction result.
Reproducibility	The documented environment, dependencies, notebooks, and processing sequence should allow experiments to be repeated.
Maintainability	Separate modules should support independent changes to data processing, modeling, evaluation, and application code.

Requirement	Description
Performance	The workflow should run on a standard development computer for the current small dataset.
Reliability	Training/testing separation and training-only preprocessing should be maintained.
Privacy	The survey is intended to be anonymous and voluntary, with no unnecessary personal identifiers.
Transparency	The application should clearly state that it is experimental and not a diagnostic tool.

3.1.2 Feasibility Study

i. Technical Feasibility

The project is technically feasible because it uses established open-source technologies including Python 3.12, pandas, NumPy, Matplotlib, Seaborn, scikit-learn, XGBoost, SHAP, Jupyter, Streamlit, Git, and GitHub. The current dataset is small enough to execute the experiments on a standard development computer.

ii. Operational Feasibility

The project is operationally feasible because survey responses can be collected using Google Forms and processed through a modular notebook workflow. The workflow separates data understanding, cleaning, EDA, feature engineering, training, evaluation, explainability, and error analysis.

iii. Economic Feasibility

The Machine Learning workflow uses open-source software and therefore does not require substantial commercial licensing costs. The main resources are a computer, internet access, development software, data-collection services, and time.

iv. Time Feasibility

The Gantt chart below represents the overall timeline of the project and the weeks in which each activity is carried out.

Table 3.2 Week Project Development Schedule (Gantt Chart) of Academic Stress Prediction

Activity	W 1	W 2	W 3	W 4	W 5	W 6	W 7	W 8	W 9	W 10	W 11	W 12	W 13	W 14	W 15	W 16
Planning & requirements																
Literature review																
Survey/data collection																
Data cleaning & preprocessing																
EDA & target construction																
Feature engineering																
Model implementation																
Cross-validation & comparison																
Held-out testing																
Hyperparameter experiment																
SHAP & error analysis																
Streamlit prototype																
Testing & validation																
Documentation																
Final review & refinement																

3.1.3 Object Modelling: Class & Object Diagram

The UML class diagram below shows the structure of the proposed system. The Classifier class is generalized into the concrete classifiers used in the project: Decision Tree, Logistic Regression, Random Forest, KNN, XGBoost, SVM, Naive Bayes and the Dummy baseline. A dashed arrow represents a dependency (uses) relationship, the line with a hollow diamond represents aggregation, and the hollow triangle represents generalization.

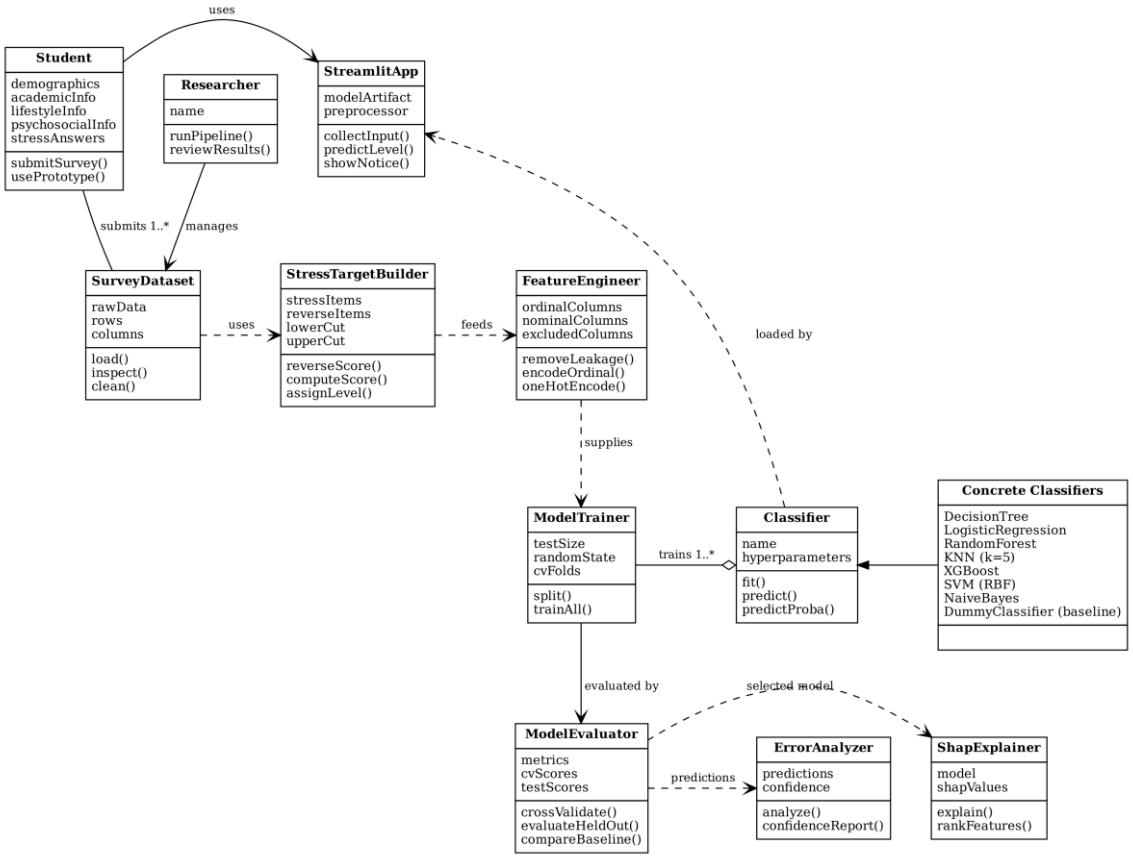


Figure 3.2 Class Diagram of Academic Stress Prediction

The Student class represents an anonymous respondent, while the Researcher class manages the dataset and runs the pipeline. The StressTargetBuilder class reverse-scores the selected items, calculates the ten-item stress score and assigns the Low, Moderate or High label; the FeatureEngineer class removes target-derived fields and encodes the variables. The ModelTrainer class trains one or more Classifier objects, the ModelEvaluator class performs cross-validation and held-out testing, and the ShapExplainer and ErrorAnalyzer classes examine the selected model. The StreamlitApp class loads the stored model to display the predicted category.

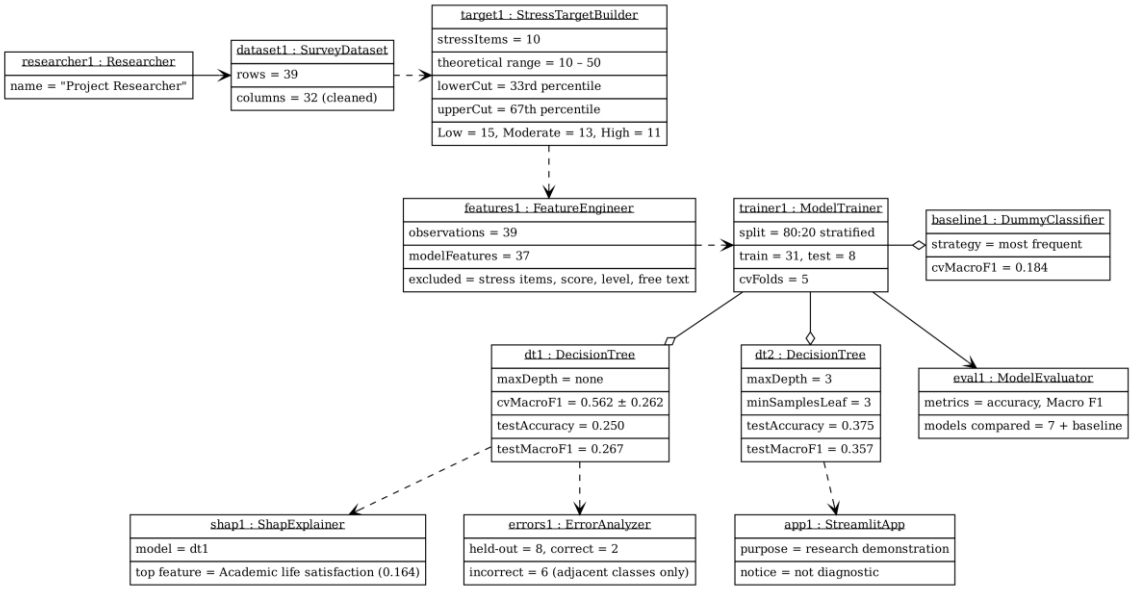


Figure 3.3 Object Diagram of Academic Stress Prediction

The object diagram above shows the classes instantiated with the values of the current experiment: 39 observations, 15 Low, 13 Moderate and 11 High observations, 37 model features, 31 training and eight held-out observations, and the original and constrained Decision Tree objects compared with the Dummy baseline.

3.1.4 Dynamic Modelling: State and Sequence Diagram

State Diagram:

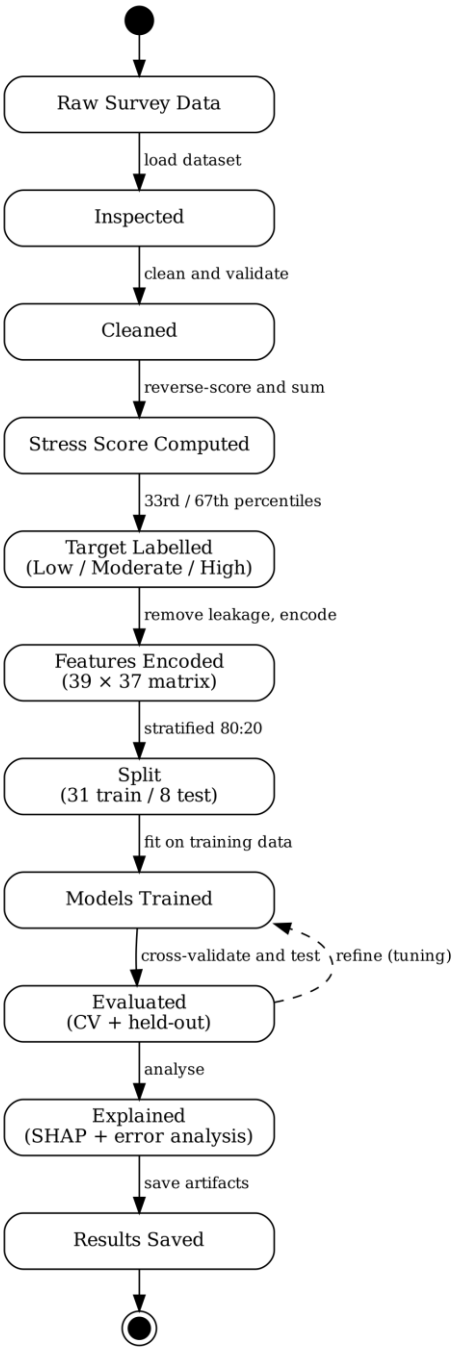


Figure 3.4 State Diagram of Academic Stress Prediction

The above state diagram illustrates the life cycle of the research data. It begins at the initial state (solid black dot) with the Raw Survey Data, which is inspected, cleaned, scored, labelled as Low, Moderate or High, encoded into the 39 × 37 feature matrix and split into 31 training and eight test observations. The models are then trained, evaluated,

explained and the results saved, which moves the process to the final state (encircled black dot).

Sequence Diagram:

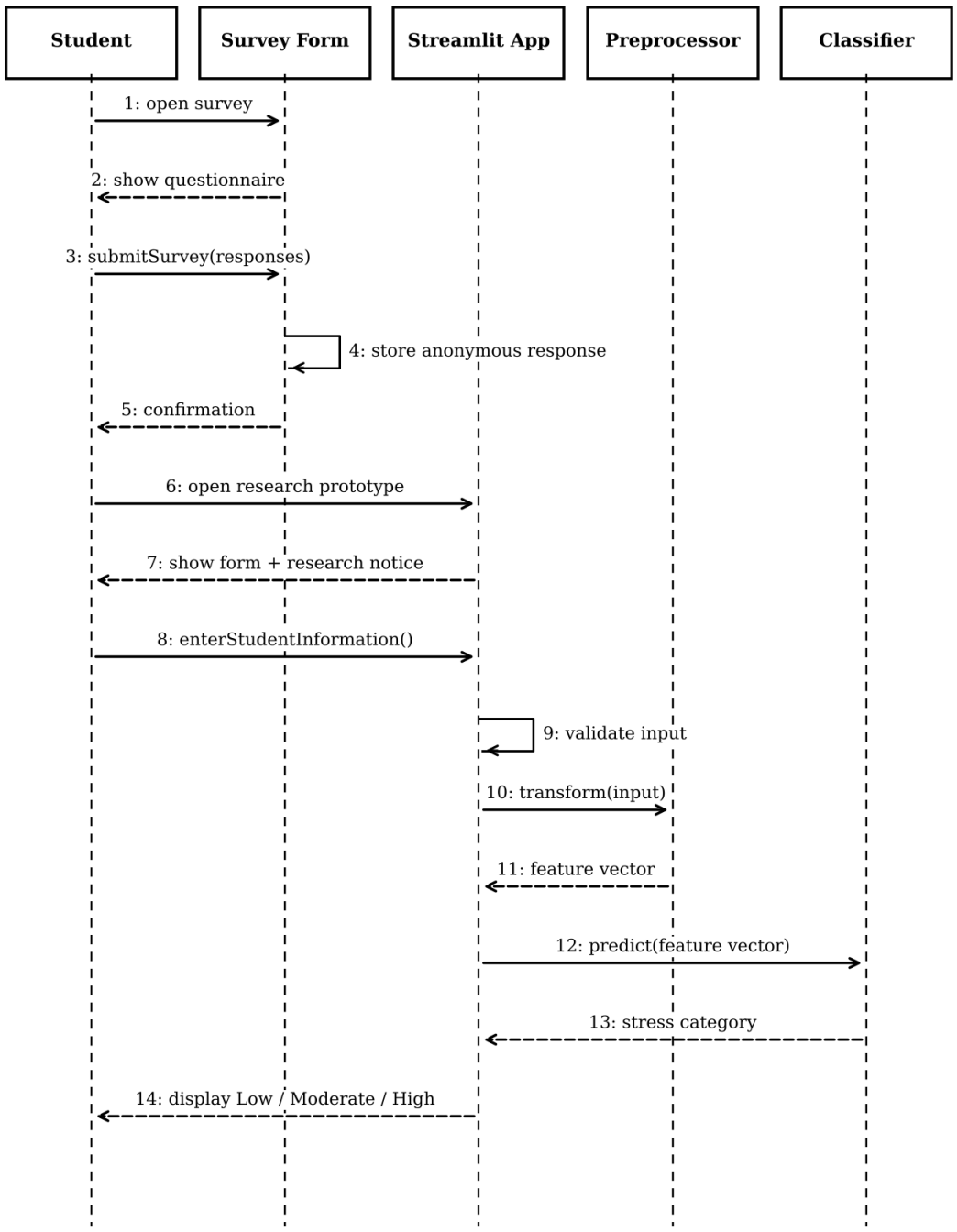


Figure 3.5 Sequence Diagram of Academic Stress Prediction - Student

The above sequence diagram shows how the student interacts with the system. The student submits the voluntary survey response, which is stored anonymously, and may then open the research prototype. The application validates the entered information, the

preprocessor returns the feature vector, the classifier returns the stress category and the application displays the Low, Moderate or High result.

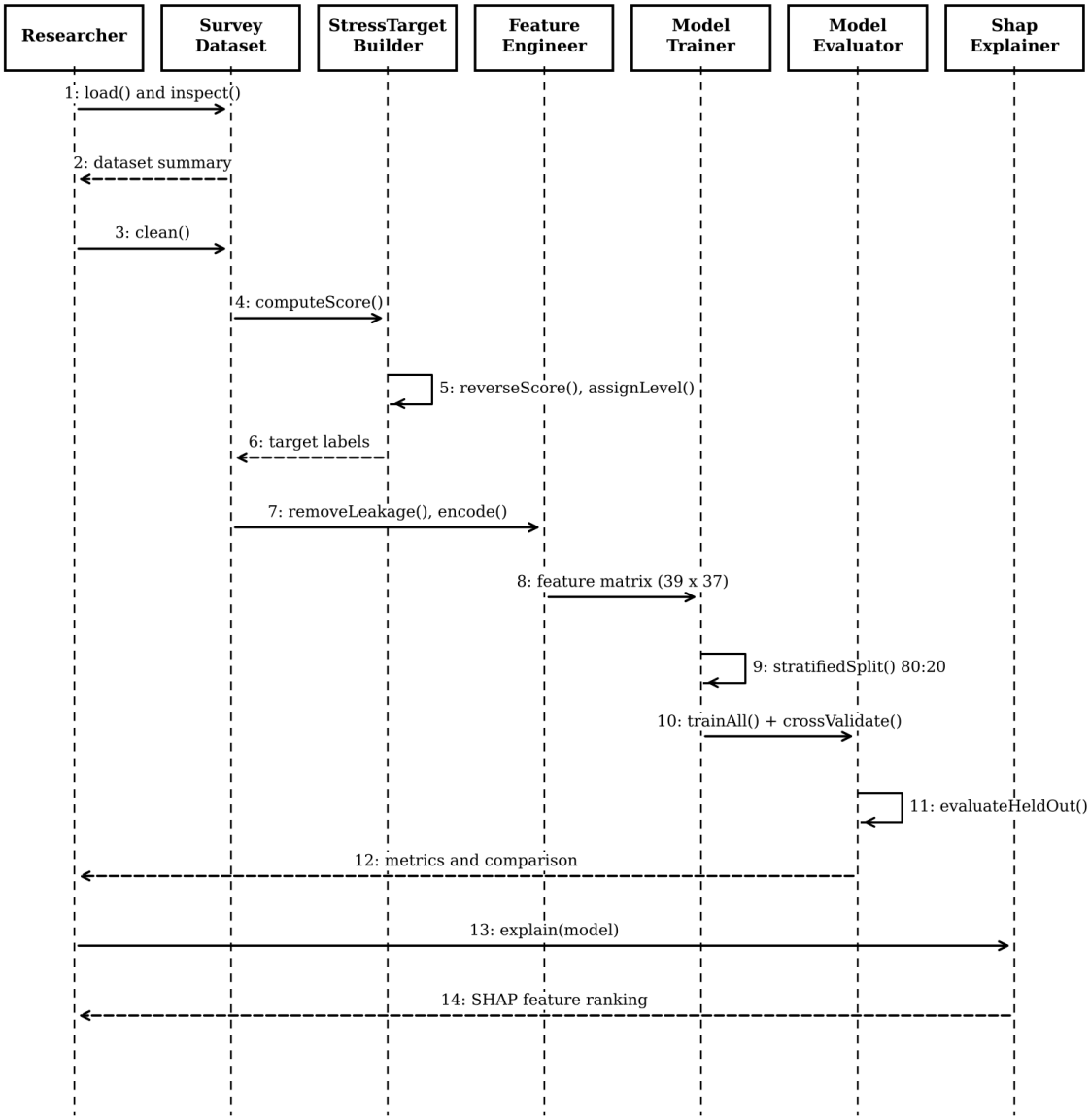


Figure 3.6 Sequence Diagram of Academic Stress Prediction - Researcher

The second sequence diagram represents the researcher's side. The researcher loads, inspects and cleans the dataset, the stress score and target are computed, the feature matrix is created and split, the classifiers are trained and evaluated, and finally the selected model is explained with SHAP.

3.1.5 Process Modelling: Activity Diagram

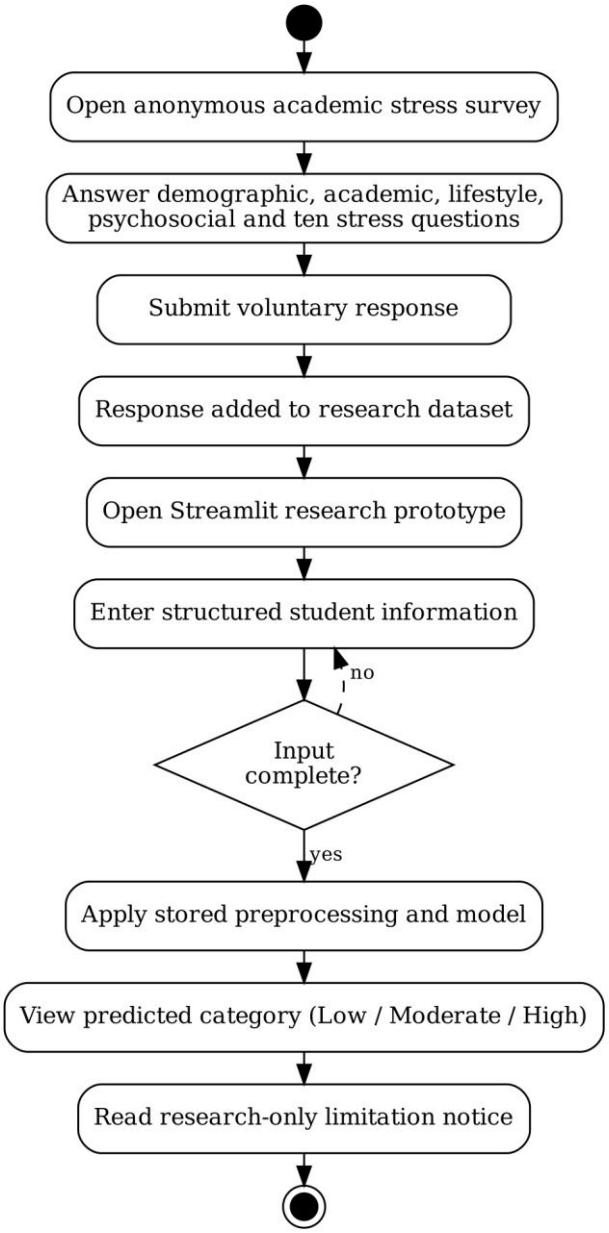


Figure 3.7 Activity Diagram of Academic Stress Prediction for Student

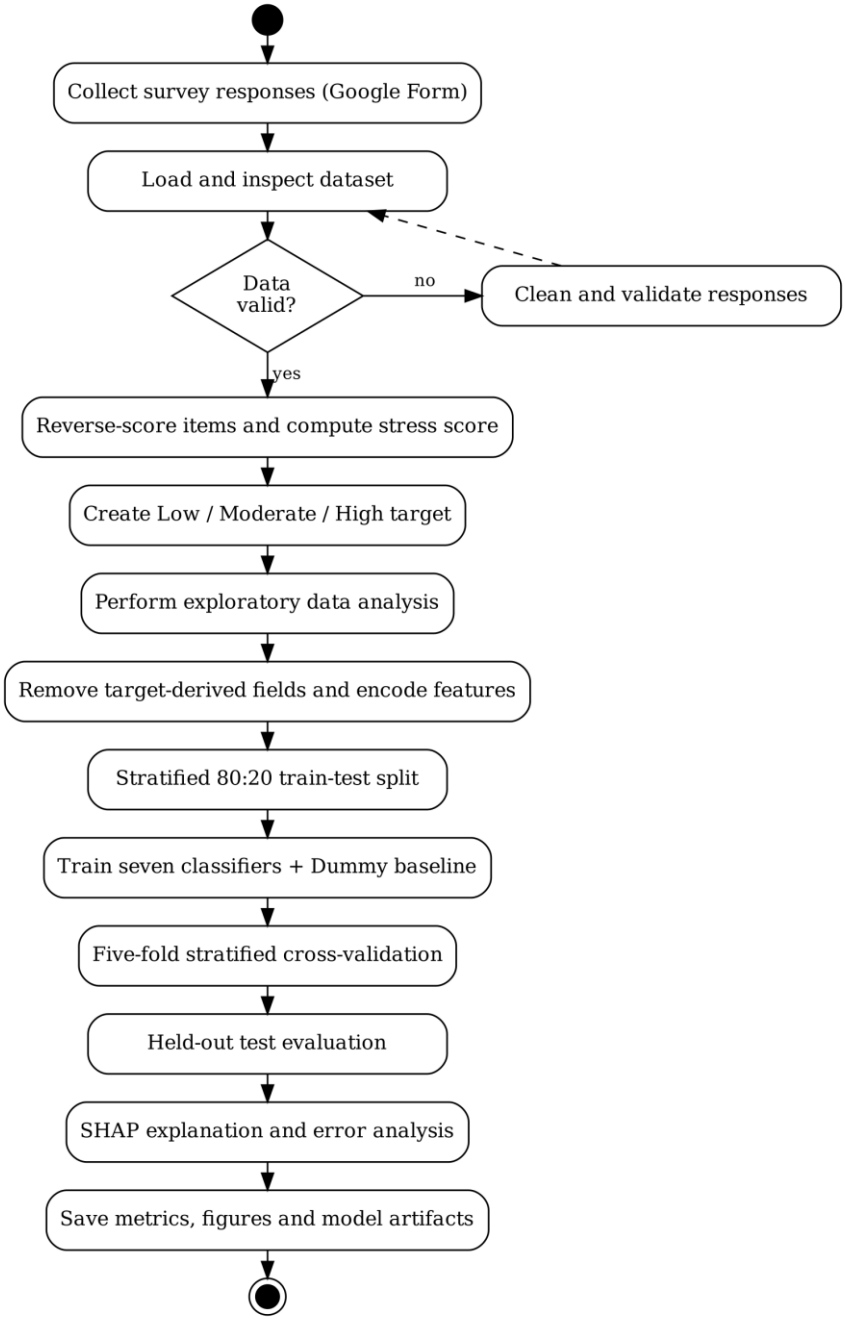


Figure 3.8 Activity Diagram of Academic Stress Prediction for Researcher

The activity diagrams above represent the processes of the student and the researcher, from the initial state (black dot) to the final state (encircled black dot). Activities are shown as rounded rectangles and decisions as diamonds. The student answers and submits the survey and may enter information in the prototype to view a predicted category; the researcher cleans and scores the data, creates the target, encodes the features, trains and evaluates the classifiers, performs SHAP and error analysis, and saves the results.

3.2 System Design

3.2.1 Refinement of Classes and Objects

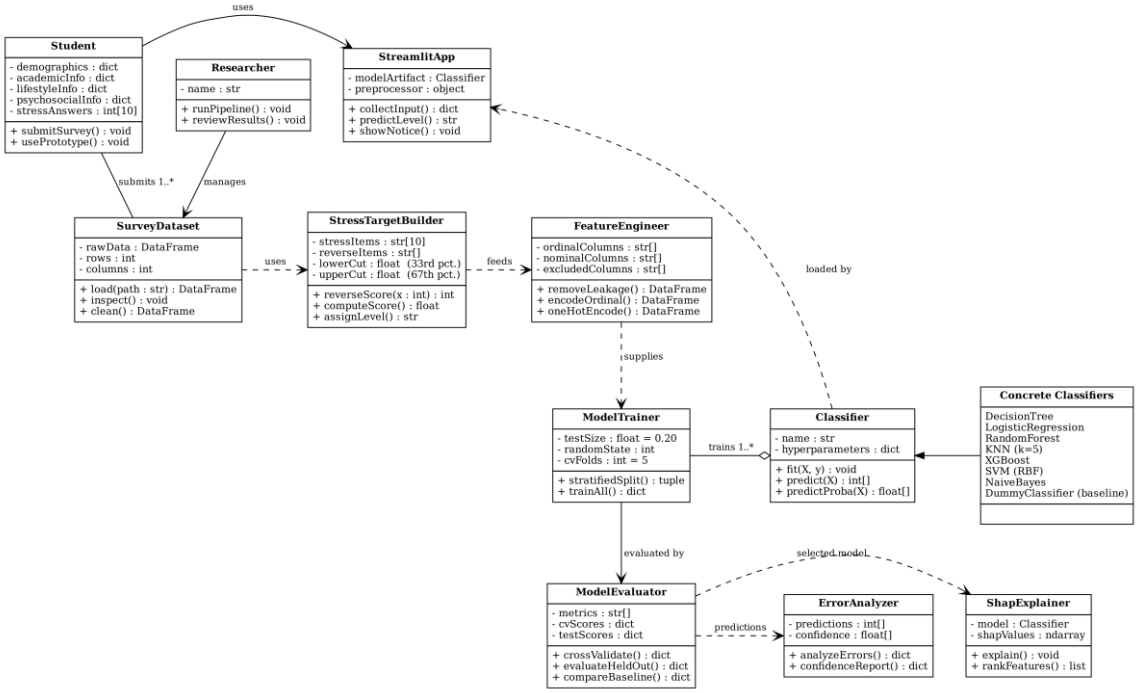


Figure 3.9 Refinement of Class Diagram of Academic Stress Prediction

In the refined class diagram, the visibility and data types of the attributes and operations are shown. The plus "+" sign denotes the public access modifier, whereas the minus "–" sign denotes the private access modifier. All attributes are private, and the operations called by other classes or by the user interface (for example submitSurvey, runPipeline, fit, predict, crossValidate and predictLevel) are public.

3.2.2 Component Diagram

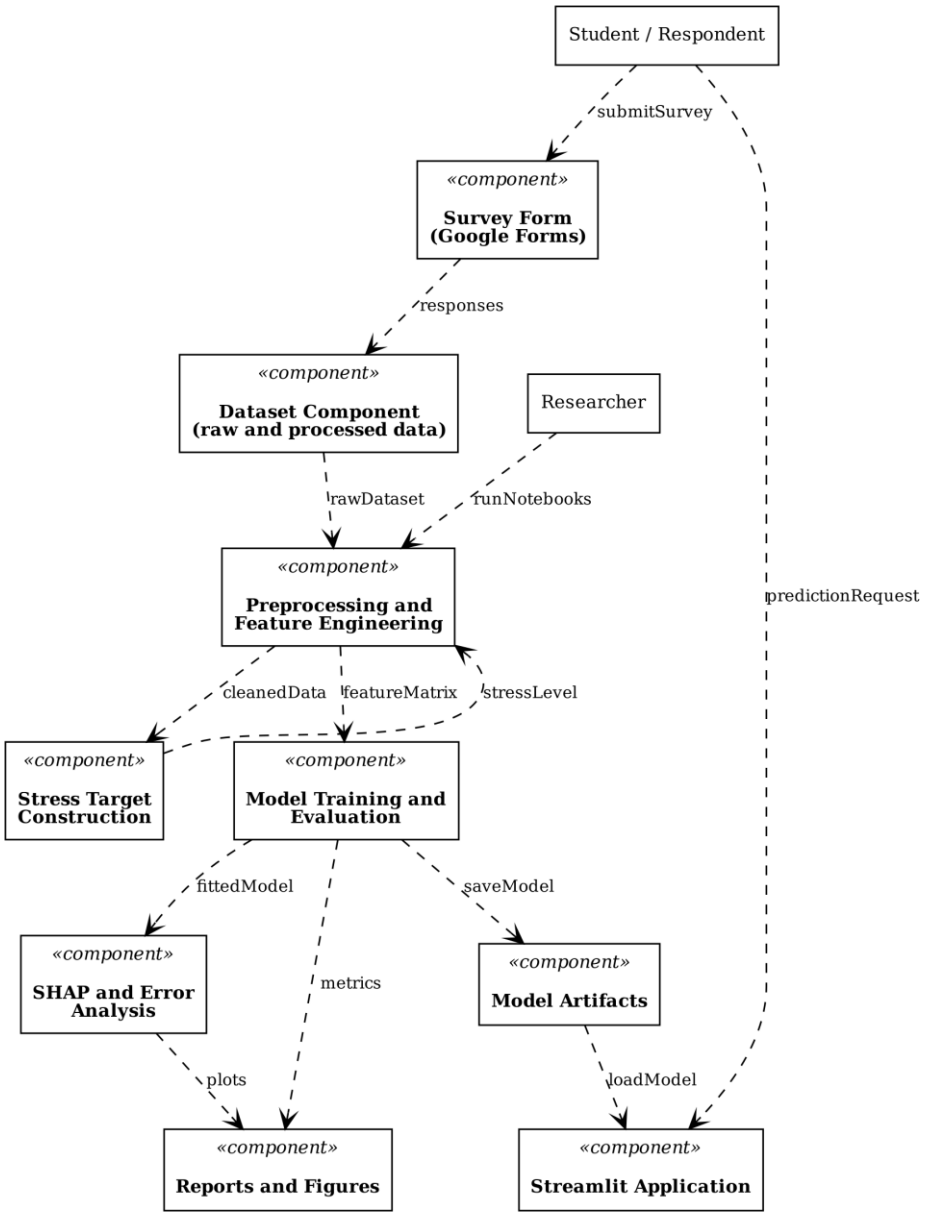


Figure 3.10 Component Diagram of Academic Stress Prediction

The component diagram above shows the nine components of the system: Survey Form (Google Forms), Dataset Component, Preprocessing and Feature Engineering, Stress Target Construction, Model Training and Evaluation, SHAP and Error Analysis, Model Artifacts, Reports and Figures, and Streamlit Application. A dashed arrow labelled with an interface name shows that one component uses the interface provided by another. The student submits the survey, the researcher runs the notebooks that prepare the data and train the models, and the Streamlit Application loads the stored model artifacts to answer prediction requests.

3.2.3 Deployment Diagram

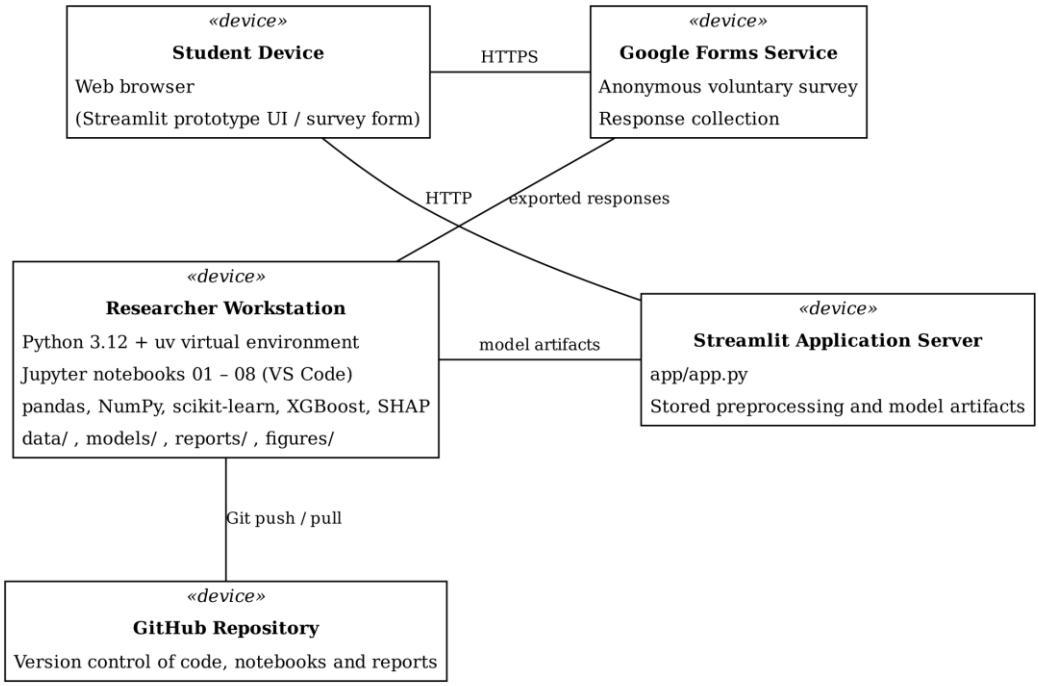


Figure 3.11 Deployment Diagram of Academic Stress Prediction

The deployment diagram above represents the nodes of the system: the Student Device, the Google Forms Service, the Researcher Workstation, the Streamlit Application Server and the GitHub Repository. The student fills the survey and uses the prototype through a web browser. The Researcher Workstation runs Python 3.12 with Jupyter notebooks and holds the data, models, reports and figures; its model artifacts are used by the Streamlit Application Server, and the source code is version-controlled with Git and GitHub.

3.2.4 System Architecture

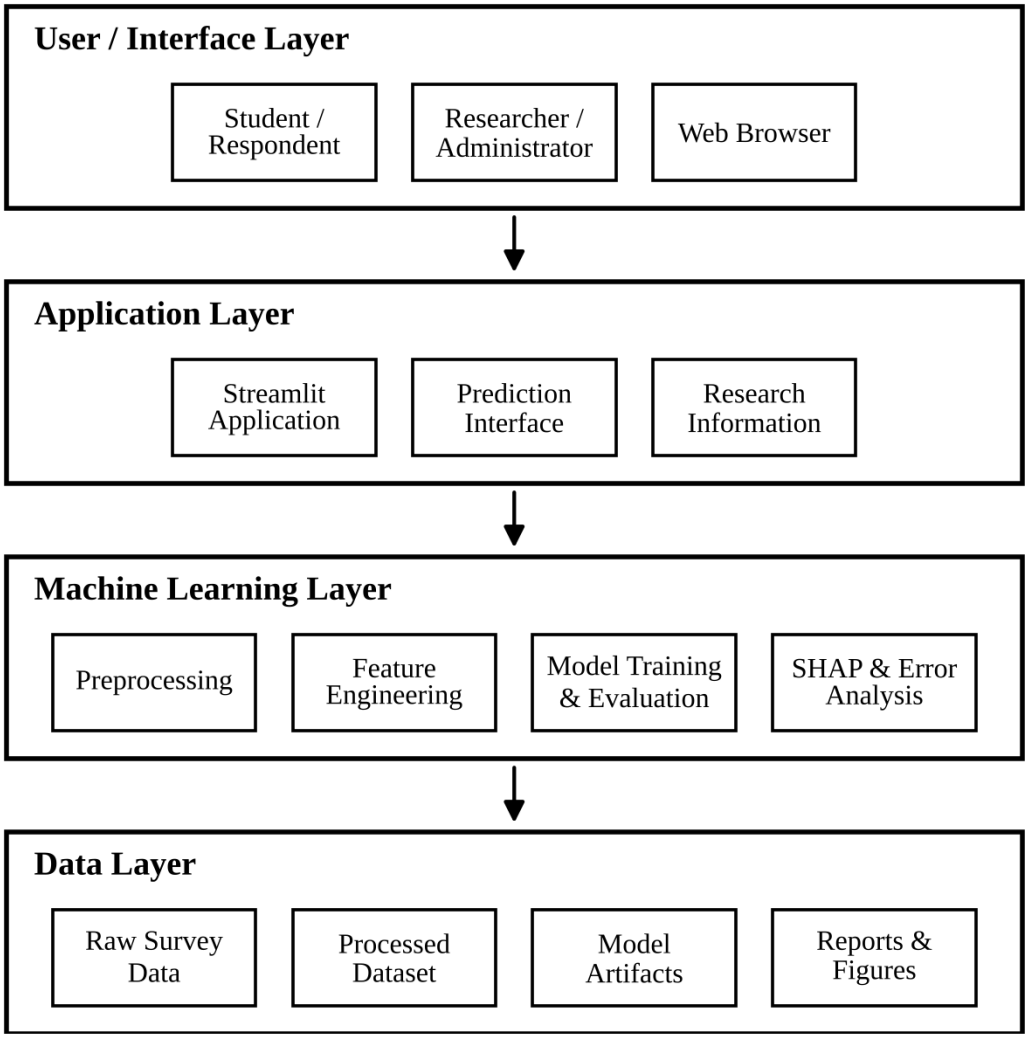


Figure 3.12 System Architecture of Academic Stress Prediction

The system architecture is organized into user/interface, application, Machine Learning, and data layers. The Streamlit layer demonstrates the prediction workflow, while notebooks implement the analytical pipeline. Data and model artifacts are stored separately from application logic.

3.2.5 Data Design

Table 3.3 Data Variables

Category	Variables
Demographic	Age, Gender, University, Program, Year, CGPA
Academic	Study hours, Attendance, Assignment stress, Exam count

Category	Variables
Lifestyle	Sleep hours, Social media usage, Physical activity, Part-time job, Screen time
Psychosocial	Financial stress, Social support, Career stress, Academic life satisfaction, Academic performance, Thought break
Stress Assessment	Ten five-point stress-assessment questions

3.2.6 Target Construction Design

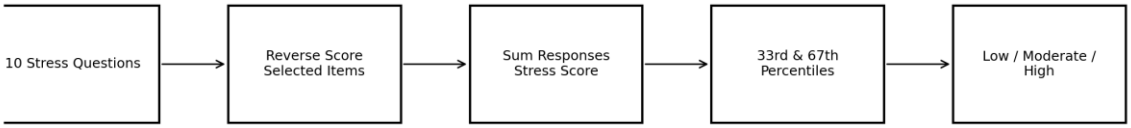


Figure 3.13 Stress Target Construction Flow

The theoretical range of the ten-question stress score is 10–50. For reverse-scored questions, the transformation is Reversed Score = 6 − Original Score. The transformed responses are summed to obtain the stress score. The 33rd and 67th percentiles of the observed distribution are then used to create Low, Moderate, and High categories.

Table 3.4 Current Stress-Level Distribution

Class	Current Count
Low	15
Moderate	13
High	11

The percentile thresholds are sample-dependent and should not be interpreted as universally established clinical thresholds.

3.3 Algorithm

The system uses classification-based algorithms for predicting the academic stress level of undergraduate students. Classification algorithms are best suited for this system because the target is a category rather than a continuous number: the stress score obtained

from the ten stress questions is converted into three classes, Low, Moderate and High. These algorithms are good at finding the relationship between the features of a student, such as study hours, sleep hours, social support, financial stress and academic life satisfaction, and the stress class. By learning how these features interact with the stress level, the models can classify new observations. Seven classification algorithms are compared with a Dummy Classifier baseline in the project. The common steps followed for all of them are given first, and the algorithms are explained afterwards.

Common Steps for All Algorithms

Step 1: Define the variables

- **Dependent variable (Target):** Academic stress level (Low, Moderate or High), created from the stress score using the 33rd and 67th percentiles.
- **Independent variables (Features):** The 37 model features obtained from the demographic, academic, lifestyle and psychosocial survey variables.

Step 2: Prepare the data

- **Data pre-processing:** Reverse-score the selected stress items, calculate the stress score, remove the target-derived fields to prevent leakage, apply ordinal encoding to ordered variables and one-hot encoding to nominal variables.
- **Train-test split:** Create a stratified 80:20 split, giving 31 training and eight held-out test observations. Where scaling is required, it is fitted only on the training data.

Step 3: Train and compare the models

- **Model training:** Fit each classifier on the training data and compare them using five-fold stratified cross-validation.
- **Baseline:** Compare each model with the Dummy Classifier, which always predicts the most frequent class.

Step 4: Evaluate the models

- **Use metric system:** Performance is evaluated using accuracy and Macro F1-score in cross-validation, and the selected model is then evaluated once on the held-out test set.

Decision Tree

A Decision Tree classifies a student by asking a sequence of simple questions about the features, such as “is academic life satisfaction below a certain value?”. It is easy to understand and interpret, which is why it is investigated further in this project using SHAP and error analysis. However, an unrestricted tree can memorize the training data and overfit.

How it works?

- **Step 1:** Start with all training observations at the root node.
- **Step 2:** Select the feature and threshold that best separates the Low, Moderate and High classes, measured by an impurity criterion such as Gini impurity.
- **Step 3:** Split the observations into branches and repeat the process for each branch.
- **Step 4:** Stop when a node is pure or a constraint is reached. In the constrained experiment, the maximum depth is 3 and the minimum samples per leaf is 3.
- **Step 5:** To predict, pass a new observation from the root to a leaf; the majority class of that leaf is the predicted stress level.

Logistic Regression

Logistic Regression is a linear classification method. For more than two classes, it calculates a score for each class as a weighted sum of the features and converts the scores into probabilities. It is used as a linear reference model.

How it works?

- **Step 1:** For each class k, calculate a score from the features: $z_k = b_k + w_{k1} \cdot x1 + w_{k2} \cdot x2 + \ldots + w_{kn} \cdot xn$.
- **Step 2:** Convert the scores into class probabilities using the softmax function:

$$P(\text{class } k \mid x) = \exp(z_k) / \Sigma \exp(z_j)$$

- **Step 3:** Learn the weights from the training data by minimizing the classification error (log-loss).
- **Step 4:** The class with the highest probability is the predicted stress level.

Random Forest

Random Forest is an ensemble of many Decision Trees. Instead of depending on a single tree, it combines the predictions of many trees, which reduces overfitting and helps represent nonlinear relationships.

How it works?

- **Step 1:** Create many different training samples by randomly sampling the training observations with replacement.
- **Step 2:** Build a Decision Tree on each sample, using only a random subset of features at each split.
- **Step 3:** For a new observation, every tree predicts a stress class.
- **Step 4:** The class receiving the most votes is the final prediction.

K-Nearest Neighbors (KNN)

KNN classifies an observation according to the classes of the most similar observations in the training data. The project uses k = 5. Because it is distance-based, the features are standardized before it is applied.

How it works?

- **Step 1:** Store the standardized training observations with their stress classes.
- **Step 2:** For a new observation, calculate the distance to every training observation.
- **Step 3:** Select the five nearest observations.
- **Step 4:** The most common class among these five neighbors is the predicted stress level.

XGBoost

XGBoost (Extreme Gradient Boosting) is a boosting method that builds decision trees one after another. Each new tree is trained to correct the mistakes made by the previous trees, and the final prediction is obtained by combining all trees.

How it works?

- **Step 1:** Start with an initial prediction for every training observation.
- **Step 2:** Calculate the errors of the current model.
- **Step 3:** Build a new small tree that predicts these errors and add it to the model with a small learning rate.

- **Step 4:** Repeat for the chosen number of trees; the combined output gives the predicted stress class.

Support Vector Machine (SVM)

SVM finds the decision boundary that separates the classes with the largest possible margin. The project uses the RBF (radial basis function) kernel, which allows nonlinear boundaries, together with standardized features.

How it works?

- **Step 1:** Standardize the features using the training data.
- **Step 2:** Use the RBF kernel to measure the similarity between two observations:

$$K(x, x') = exp(-\gamma \|x - x'\|^2)$$

- **Step 3:** Find the boundary that maximizes the margin between the classes, using the training observations closest to the boundary (support vectors).
- **Step 4:** A new observation is assigned the class on the side of the boundary where it falls.

Naive Bayes

Naive Bayes is a probabilistic classifier based on Bayes' theorem. It assumes that the features are conditionally independent given the class, which makes it simple and fast.

How it works?

- **Step 1:** Calculate the prior probability of each stress class from the training data.
- **Step 2:** Estimate the probability of each feature value within each class.
- **Step 3:** For a new observation, combine them using Bayes' theorem:

$$P(class \mid x) \propto P(class) \times P(x1 \mid class) \times P(x2 \mid class) \times \ldots \times P(xn \mid class)$$

- **Step 4:** The class with the highest probability is the predicted stress level.

CHAPTER 4
IMPLEMENTATION AND TESTING

4.1 Implementation

4.1.1 Tools Used

The project is built with Python and its Machine Learning libraries, and the prototype is presented with Streamlit. The tools used are summarized below.

Table 4.1 Development Environment

Component	Specification / Technology
Programming Language	Python 3.12
Environment	uv + Python virtual environment
Data Processing	pandas, NumPy
Visualization	Matplotlib, Seaborn
Machine Learning	scikit-learn
Boosting	XGBoost
Explainability	SHAP
Application	Streamlit
Notebook Environment	Jupyter
Editor	Visual Studio Code
File/Data Support	openpyxl
Version Control	Git / GitHub

The experiments were designed for a standard development computer. Because the current dataset contains only 39 observations, the computational requirements are modest.

4.1.2 Implementation Details of Modules

a. Data Collection Module

The project uses an anonymous voluntary Google Form-based academic stress survey. The questionnaire contains demographic, academic, lifestyle, psychosocial, and stress-assessment questions. The current cleaned dataset contains 39 observations and 32 cleaned columns before the final model feature transformation.

b. Data Preprocessing Module

The preprocessing stage first examines dataset dimensions, data types, missing values, category representations, and distributions. Cleaning and transformation are then applied to prepare the dataset for stress-score construction and Machine Learning.

The ten stress questions use a five-point scale from Never (1) to Very Often (5). Selected items are reverse-scored using Reversed Score = 6 − Original Score. The total stress score is obtained by summing the transformed responses.

Table 4.2 Stress Score Summary

Statistic	**Value**
Number of stress questions	10
Theoretical minimum	10
Theoretical maximum	50
Observed minimum	10
Observed maximum	39
Observed mean	28.95
Observed standard deviation	4.82

The target is created from the 33rd and 67th percentiles of the observed stress-score distribution. The resulting current class counts are 15 Low, 13 Moderate, and 11 High observations.

c. Feature Engineering Module

The final model matrix contains 37 features. Stress questions and target-derived variables are excluded to prevent leakage. Ordered survey variables are ordinal encoded and nominal categorical variables are one-hot encoded.

The free-text biggest_stress_reason variable is excluded because the current project does not implement NLP. This design keeps the experiment focused on structured survey information.

d. Model Training Module

Seven classification algorithms are trained using the same general feature-engineering workflow. The models are Decision Tree, Logistic Regression, Random Forest, KNN, XGBoost, SVM, and Naive Bayes. A Dummy Classifier is used as a baseline.

The data are divided into 31 training observations and eight held-out test observations using an 80:20 stratified split. Standardization, where required, is fitted only on training data.

e. Streamlit Prototype Module

A Streamlit-based application is included as a research demonstration. The planned interface accepts relevant structured student inputs, applies the stored preprocessing and model artifacts, displays a predicted Low, Moderate, or High category, and presents a prominent research limitation notice.

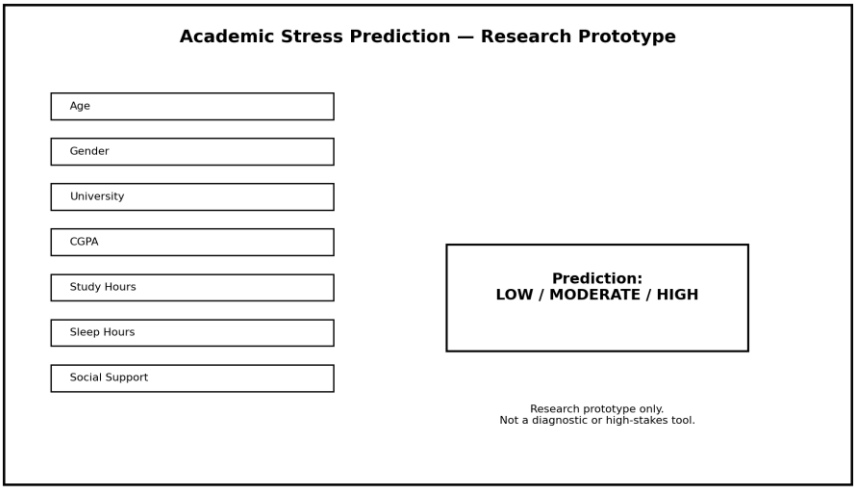


Figure 4.1 Streamlit Prototype Interface Design

The prototype is not a diagnostic system, mental-health assessment tool, or high-stakes decision system. Its purpose is to demonstrate how the Machine Learning pipeline could be exposed through an interactive interface.

4.2 Testing

4.2.1 Test Case for Unit Testing

Table 4.3 Test Case for Data Preparation

Test Case Id	Description	Input Test Data	Expected Result	Actual Result	Pass/Fail

Test Case Id	Description	Input Test Data	Expected Result	Actual Result	Pass/Fail
TC1.1	Reverse scoring of selected stress items	Response x = 2 on a reverse-scored item	Selected item x becomes 6 − x (i.e., 4)	Transformation implemented	Pass
TC1.2	Stress score calculation	Ten transformed stress items (scale 1–5)	Items sum correctly (range 10–50)	Stress score generated (observed range 10–39)	Pass
TC1.3	Target construction	Stress scores and 33rd / 67th percentile thresholds	Low, Moderate and High categories created	15 Low, 13 Moderate, 11 High	Pass
TC1.4	Feature exclusion (leakage prevention)	Stress questions, reverse-scored fields, stress score, stress level	Target-derived fields excluded	Leakage fields removed	Pass
TC1.5	Encoding of variables	Ordinal and nominal variables	Variables transformed	Final matrix created (39 × 37)	Pass

4.2.2 Test Case for System Testing

Table 4.4 Black Box System Testing

S.N.	Description	Requirements	Result	Test Status
1	End-to-end notebook sequence	Survey dataset and notebooks 01 to 08 executed in order	All stages execute in order	Passed

S.N.	Description	Requirements	Result	Test Status
2	Model comparison	Feature matrix, seven classifiers and the Dummy baseline	Cross-validation and held-out metrics generated	Passed
3	Streamlit prototype	Stored model artifacts; student input	Application loads the model and supports demonstration; under development	Partial

4.3 Result Analysis

4.3.1 Cross-Validation Results

Table 4.5 Classification Algorithm Comparison (Cross-Validation and Held-Out Results)

Model	CV Accuracy	CV Macro F1	Held-Out Accuracy	Held-Out Macro F1
Decision Tree	0.605 ± 0.231	0.562 ± 0.262	0.250	0.267
Logistic Regression	0.414 ± 0.146	0.389 ± 0.161	—	—
Random Forest	0.410 ± 0.185	0.364 ± 0.197	—	—
KNN	0.381 ± 0.142	0.342 ± 0.163	—	—
XGBoost	0.381 ± 0.142	0.323 ± 0.159	—	—
SVM	0.376 ± 0.209	0.292 ± 0.223	—	—
Naive Bayes	0.286 ± 0.181	0.291 ± 0.181	—	—
Dummy Classifier	0.386	0.184	0.386	0.184

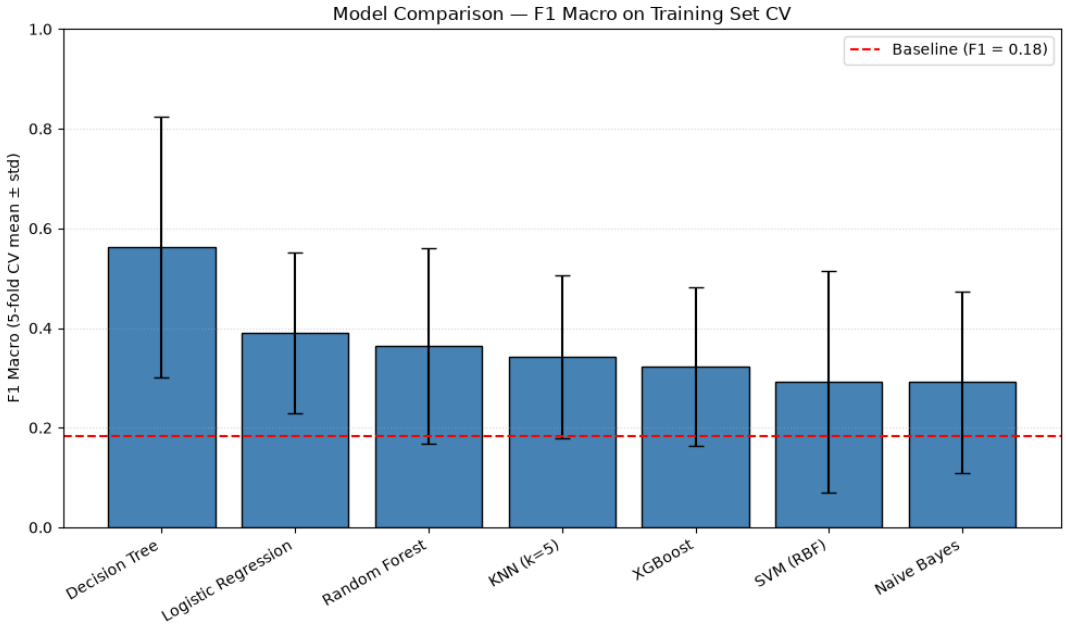


Figure 4.2 Five-Fold Cross-Validation Macro F1-Score

The Decision Tree has the highest cross-validation Macro F1-score among the tested Machine Learning models at 0.562 ± 0.262. However, the standard deviation is large relative to the mean, reflecting instability in the small validation folds.

4.3.2 Held-Out Test Results

Table 4.6 Held-Out Test Results of the Decision Tree and Dummy Baseline

Metric	Decision Tree	Dummy Baseline
Test observations	8	8
Accuracy	0.250	0.386
Macro F1	0.267	0.184

The held-out Decision Tree results are substantially lower than the cross-validation results. The model correctly classifies only two of the eight held-out observations. The test accuracy of 0.250 is also below the Dummy Classifier baseline accuracy of 0.386.

Because the test set contains only eight observations, each individual prediction changes accuracy by 12.5 percentage points. Therefore, the held-out metrics should be interpreted as an indication of instability rather than a stable estimate of real-world performance.

4.3.3 Decision Tree Constrained Experiment

A constrained Decision Tree experiment was performed using a maximum depth of 3 and a minimum samples-per-leaf constraint of 3. The constrained tree achieved test accuracy of 0.375 and Macro F1 of 0.357, compared with 0.250 accuracy and 0.267 Macro F1 for the original tree.

Although the constrained tree performs better on the eight held-out observations, the sample remains too small to establish that the constraint improves generalization beyond this particular split.

Table 4.7 Original and Constrained Decision Tree on the Held-Out Test Set

Configuration	Test accuracy	Test Macro F1	Interpretation
Original Decision Tree	0.250	0.267	Below baseline accuracy; weak test performance
Constrained Decision Tree (depth=3, leaf=3)	0.375	0.357	Improved on this split; still preliminary

4.3.4 SHAP Explainability

Table 4.8 SHAP Feature Importance

Rank	Feature	Mean |SHAP|
1	Academic life satisfaction	0.164
2	Social support	0.145
3	Gender (Male)	0.120
4	Academic performance	0.107
5	CGPA	0.058
6	University (Sudurpaschim)	0.037
7	Physical activity	0.032

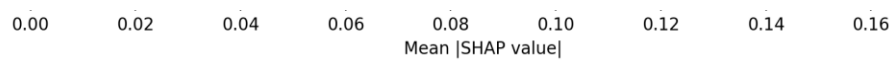


Figure 4.3 Exploratory SHAP Feature Importance

Academic life satisfaction, social support, gender, academic performance, and CGPA appear among the features with comparatively larger mean absolute SHAP values in the current Decision Tree experiment.

The SHAP analysis is exploratory. Some feature directions may appear counterintuitive in individual explanations, and the project documentation specifically warns that these results may reflect overfitting and the very small dataset. SHAP values should not be interpreted as causal effects.

4.3.5 Error Analysis

Table 4.9 Error Analysis of Held-Out Predictions

Error Analysis Item	**Observed Result**
Held-out observations	8
Correct predictions	2
Incorrect predictions	6
Error pattern	All observed errors were between adjacent stress classes (Low↔Moderate or Moderate↔High).
Low↔High errors	None observed
Mean confidence: correct predictions	1.000
Mean confidence: incorrect predictions	0.945

The six incorrect predictions occurred between adjacent classes. No Low-to-High or High-to-Low error was observed. Because the test set contains only eight observations, this pattern is descriptive of the current experiment and cannot be generalized.

The model was also overconfident on incorrect cases: mean confidence was 0.945 for incorrect predictions compared with 1.000 for correct predictions. This indicates that high model confidence does not guarantee correct classification in the current small-sample setting.

4.3.6 Discussion of Results

The current experiment demonstrates that a complete survey-to-model pipeline can be implemented using structured student information. The Decision Tree records the highest

cross-validation Macro F1 among the tested models, but its held-out performance falls considerably. The difference between internal validation and held-out evaluation is consistent with the project's stated limitation of having very few observations relative to the number of features.

The results therefore should not be used to claim that the system can reliably predict academic stress among undergraduate students in Nepal. Instead, they demonstrate the importance of independent evaluation, baseline comparison, error analysis, and explicit documentation of sample limitations.

CHAPTER 5
CONCLUSION AND FUTURE RECOMMENDATION

5.1 Conclusion

This project developed a survey-based Machine Learning research prototype for investigating academic stress among undergraduate students in Nepal. The system integrates survey data collection, preprocessing, stress-score construction, target classification, feature engineering, Machine Learning model comparison, evaluation, explainability, error analysis, and an interactive Streamlit demonstration.

The current dataset contains 39 observations and 37 final model features. The ten-question stress assessment produces a numerical score that is converted into Low, Moderate, and High categories using observed percentile thresholds. Seven classification algorithms were compared with a Dummy Classifier baseline using five-fold stratified cross-validation and a held-out test set.

The Decision Tree produced the highest initial cross-validation Macro F1-score of 0.562 ± 0.262 among the tested Machine Learning models. However, on the eight held-out test observations, the original Decision Tree achieved only 0.250 accuracy and 0.267 Macro F1. A constrained-tree hyperparameter experiment (maximum depth = 3; minimum samples per leaf = 3) improved the same split to 0.375 accuracy and 0.357 Macro F1. This is a modest observed improvement, but the test set is too small to establish reliable generalization. Class-wise recall and precision must also be reported from the saved test predictions before making any stronger performance claim.

Overall, the project establishes an end-to-end experimental framework, but the current real-data evidence is insufficient for deployment or population-level prediction. The main value of the implementation is as a research prototype and a foundation for future collection and independent validation using a larger, diverse, genuine dataset.

5.2 Future Recommendations

Priority improvements include collecting several hundred genuine and more diverse responses; exporting class-wise precision, recall, F1-score, support, macro/weighted averages and confusion matrices; conducting systematic hyperparameter search using training-only cross-validation; and evaluating the final selected model once on a held-out

test set. Comparisons with published studies must document dataset size, target construction, validation protocol, and metrics.

The primary future goal is to collect approximately 300–500 genuine survey responses from a more diverse student population and re-run the same pipeline, which would support a stronger claim about generalization. Further improvements include feature selection, external validation, analysis of the free-text stress reason using NLP, and completing and testing the Streamlit prototype.

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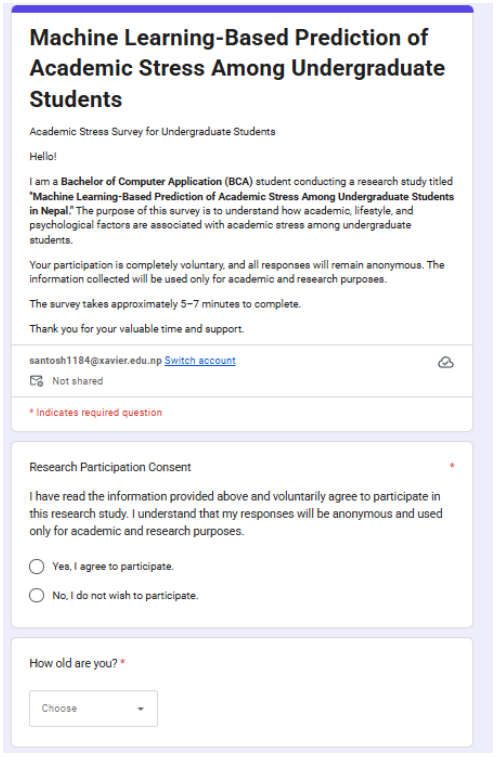
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APPENDICES

Appendix A: Survey Questionnaire

Academic Stress Survey for Undergraduate Students Form: [Form Link Here]



Machine Learning-Based Prediction of Academic Stress Among Undergraduate Students

Academic Stress Survey for Undergraduate Students

Hello!

I am a **Bachelor of Computer Application (BCA)** student conducting a research study titled **"Machine Learning-Based Prediction of Academic Stress Among Undergraduate Students in Nepal."** The purpose of this survey is to understand how academic, lifestyle, and psychological factors are associated with academic stress among undergraduate students.

Your participation is completely voluntary, and all responses will remain anonymous. The information collected will be used only for academic and research purposes.

The survey takes approximately 5–7 minutes to complete.

Thank you for your valuable time and support.

santosh1184@xavier.edu.np Switch account

Not shared

* Indicates required question

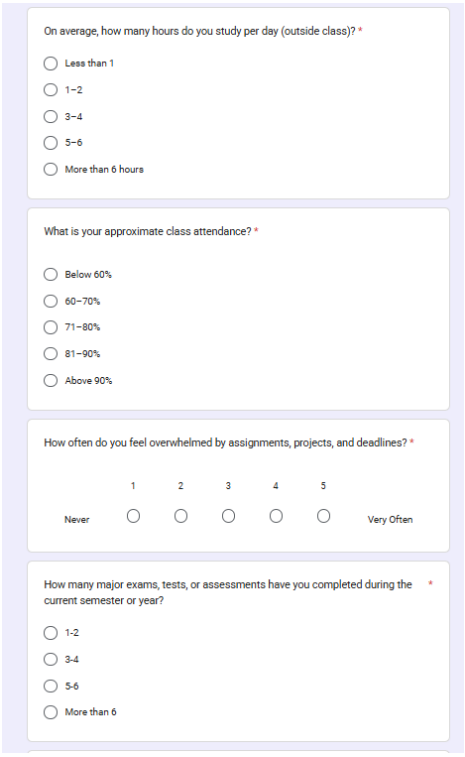
Research Participation Consent *

I have read the information provided above and voluntarily agree to participate in this research study. I understand that my responses will be anonymous and used only for academic and research purposes.

- Yes, I agree to participate.
- No, I do not wish to participate.

How old are you? *

Choose



On average, how many hours do you study per day (outside class)? *

- Less than 1
- 1–2
- 3–4
- 5–6
- More than 6 hours

What is your approximate class attendance? *

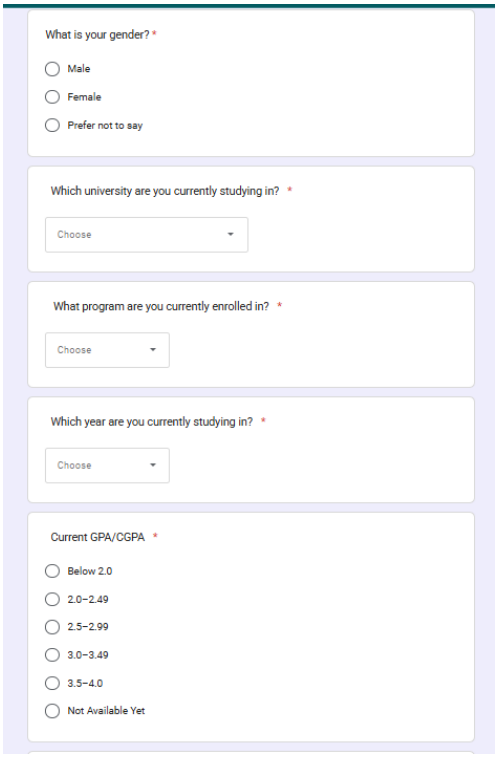
- Below 60%
- 60–70%
- 71–80%
- 81–90%
- Above 90%

How often do you feel overwhelmed by assignments, projects, and deadlines? *

	1	2	3	4	5	
Never	○	○	○	○	○	Very Often

How many major exams, tests, or assessments have you completed during the current semester or year? *

- 1-2
- 3-4
- 5-6
- More than 6



What is your gender? *

- Male
- Female
- Prefer not to say

Which university are you currently studying in? *

Choose

What program are you currently enrolled in? *

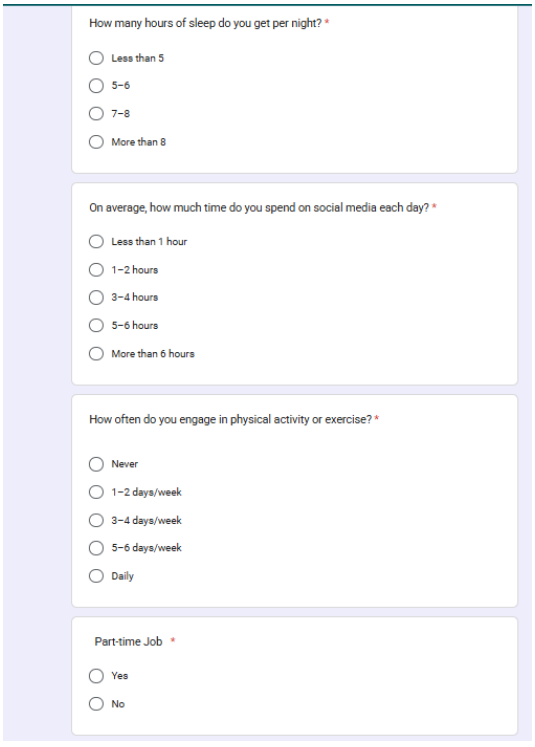
Choose

Which year are you currently studying in? *

Choose

Current GPA/CGPA *

- Below 2.0
- 2.0–2.49
- 2.5–2.99
- 3.0–3.49
- 3.5–4.0
- Not Available Yet



How many hours of sleep do you get per night? *

- Less than 5
- 5–6
- 7–8
- More than 8

On average, how much time do you spend on social media each day? *

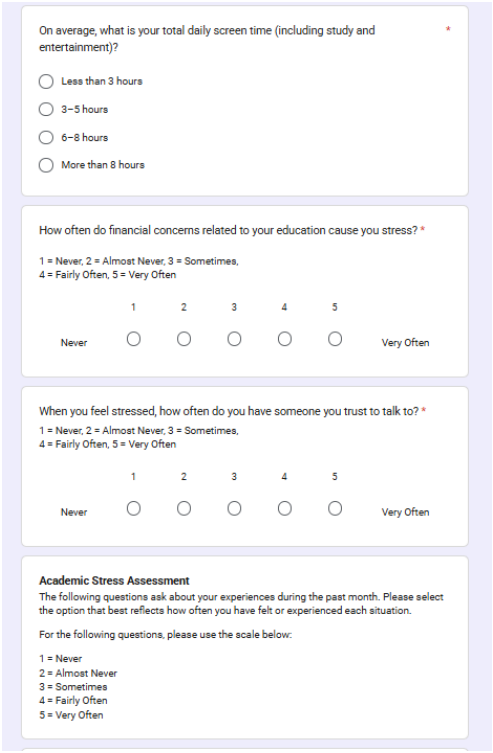
- Less than 1 hour
- 1–2 hours
- 3–4 hours
- 5–6 hours
- More than 6 hours

How often do you engage in physical activity or exercise? *

- Never
- 1–2 days/week
- 3–4 days/week
- 5–6 days/week
- Daily

Part-time Job *

- Yes
- No



On average, what is your total daily screen time (including study and entertainment)? *

- Less than 3 hours
- 3–5 hours
- 6–8 hours
- More than 8 hours

How often do financial concerns related to your education cause you stress? *

1 = Never, 2 = Almost Never, 3 = Sometimes,
4 = Fairly Often, 5 = Very Often

	1	2	3	4	5	
Never	○	○	○	○	○	Very Often

When you feel stressed, how often do you have someone you trust to talk to? *

1 = Never, 2 = Almost Never, 3 = Sometimes,
4 = Fairly Often, 5 = Very Often

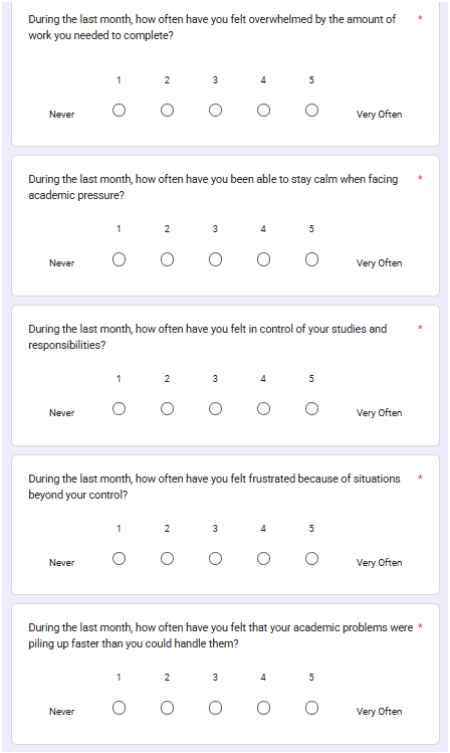
	1	2	3	4	5	
Never	○	○	○	○	○	Very Often

Academic Stress Assessment

The following questions ask about your experiences during the past month. Please select the option that best reflects how often you have felt or experienced each situation.

For the following questions, please use the scale below:

1 = Never
2 = Almost Never
3 = Sometimes
4 = Fairly Often
5 = Very Often



During the last month, how often have you felt overwhelmed by the amount of work you needed to complete? *

	1	2	3	4	5	
Never	○	○	○	○	○	Very Often

During the last month, how often have you been able to stay calm when facing academic pressure? *

	1	2	3	4	5	
Never	○	○	○	○	○	Very Often

During the last month, how often have you felt in control of your studies and responsibilities? *

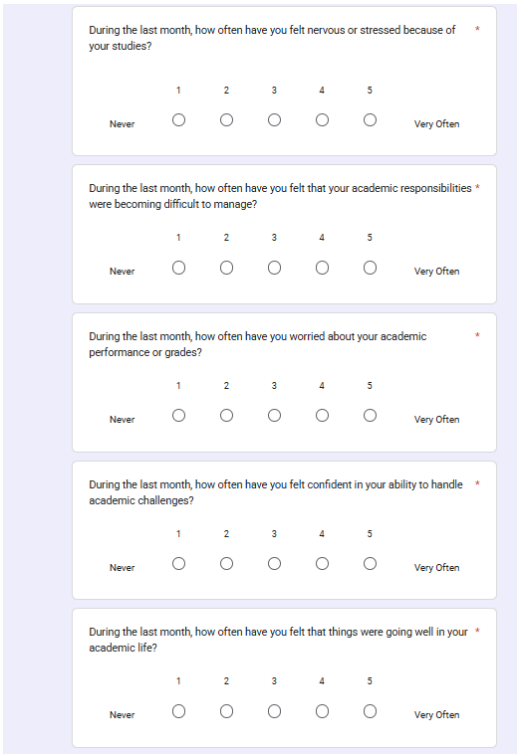
	1	2	3	4	5	
Never	○	○	○	○	○	Very Often

During the last month, how often have you felt frustrated because of situations beyond your control? *

	1	2	3	4	5	
Never	○	○	○	○	○	Very Often

During the last month, how often have you felt that your academic problems were piling up faster than you could handle them? *

	1	2	3	4	5	
Never	○	○	○	○	○	Very Often



During the last month, how often have you felt nervous or stressed because of your studies? *

	1	2	3	4	5	
Never	○	○	○	○	○	Very Often

During the last month, how often have you felt that your academic responsibilities were becoming difficult to manage? *

	1	2	3	4	5	
Never	○	○	○	○	○	Very Often

During the last month, how often have you worried about your academic performance or grades? *

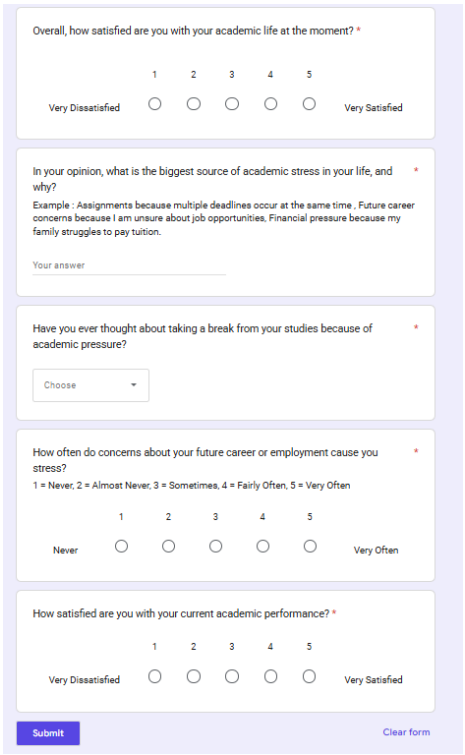
	1	2	3	4	5	
Never	○	○	○	○	○	Very Often

During the last month, how often have you felt confident in your ability to handle academic challenges? *

	1	2	3	4	5	
Never	○	○	○	○	○	Very Often

During the last month, how often have you felt that things were going well in your academic life? *

	1	2	3	4	5	
Never	○	○	○	○	○	Very Often



Overall, how satisfied are you with your academic life at the moment? *

	1	2	3	4	5	
Very Dissatisfied	○	○	○	○	○	Very Satisfied

In your opinion, what is the biggest source of academic stress in your life, and why? *

Example : Assignments because multiple deadlines occur at the same time , Future career concerns because I am unsure about job opportunities, Financial pressure because my family struggles to pay tuition.

Your answer

Have you ever thought about taking a break from your studies because of academic pressure? *

Choose

How often do concerns about your future career or employment cause you stress? *

1 = Never, 2 = Almost Never, 3 = Sometimes, 4 = Fairly Often, 5 = Very Often

	1	2	3	4	5	
Never	○	○	○	○	○	Very Often

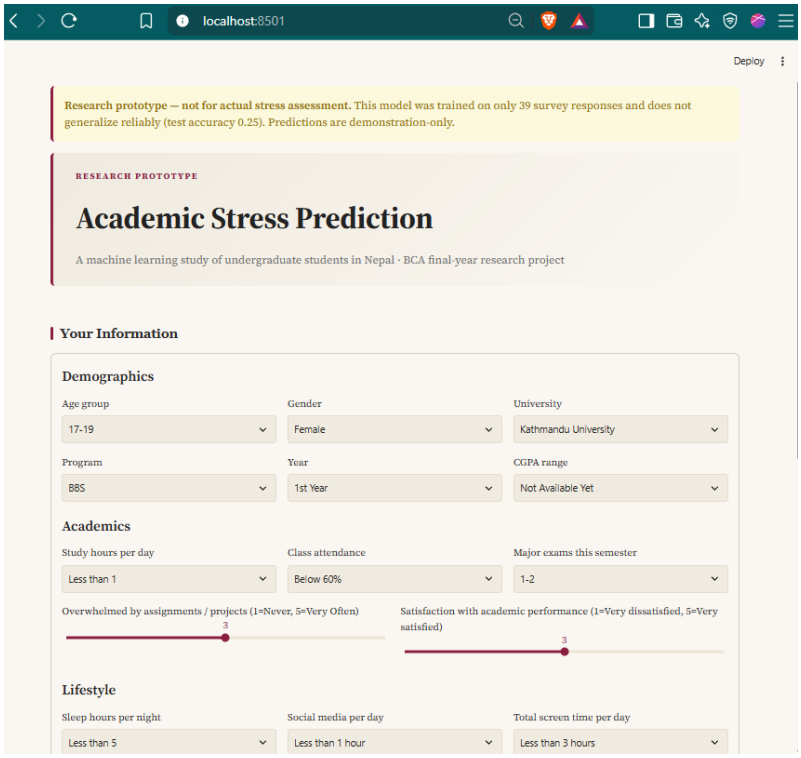
How satisfied are you with your current academic performance? *

	1	2	3	4	5	
Very Dissatisfied	○	○	○	○	○	Very Satisfied

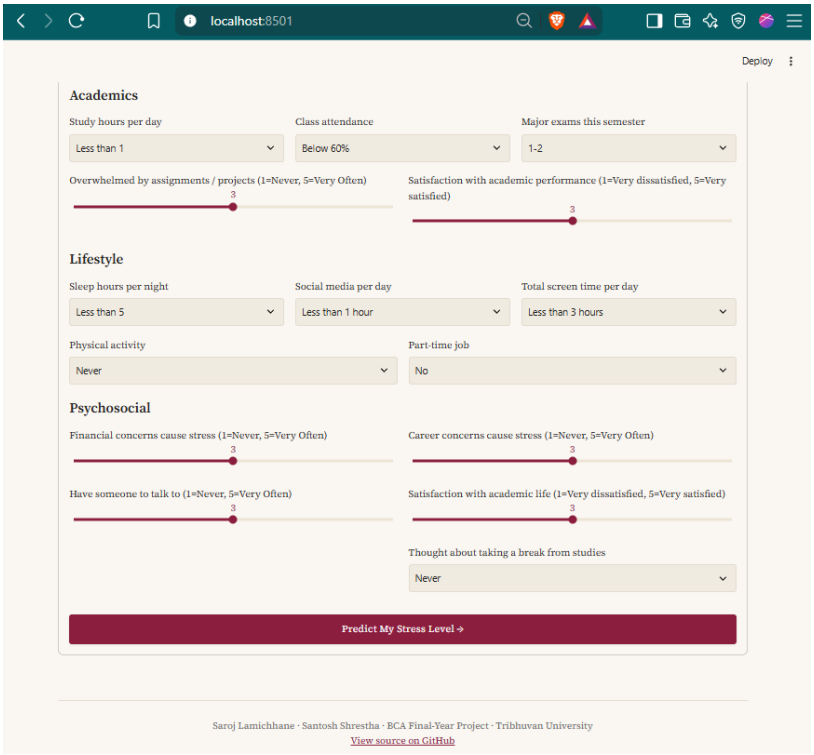
Submit Clear form

Appendix B: System Screenshots

Streamlit Prototype - Student Information Form



Streamlit Prototype - Lifestyle and Psychosocial Inputs



Streamlit Prototype - Prediction Result

